fatty acids per 100 g of total lipid are converted to g fatty acids per 100 g of food, values of less than 0.0005 are rounded to 0.

Cholesterol. Cholesterol values are generated primarily by gas liquid chromatographic procedures (AOAC 994.10). Recent meat data has been determined by a GC method without derivitization (Dinh *et al*., 2008). It is assumed that cholesterol is present only in foods of animal origin and foods containing at least one ingredient of animal origin (for example, cake that contains eggs). For mixtures containing ingredients derived from animal products, the cholesterol value may be calculated from the value for those ingredients. For foods that contain only plant products, the value for cholesterol is assumed to be zero.

Plant sterols. Data on plant sterols (campesterol, stigmasterol, and β-sitosterol) are obtained by gas-chromatographic procedures (AOAC 967.18) and summed to calculate total phytosterols (Nutr. No. 636). Plant sterols for a number of nuts, seeds, mushrooms, and other food items were determined by a gas-chromatographic method developed by Phillips *et al*. (2005) which includes an acid hydrolysis step. These data include additional sterols such as ergosterol or delta-5-avenasterol and various stanols plus some minor sterols that are not disseminated in SR. When available, data on these phytosterols are provided in a footnote for the specific food item. In these cases, Nutrient No. 636, total phytosterols, is not disseminated for these food items.

***Amino Acids*.** Amino acid data for a class or species of food are aggregated to yield a set of values that serve as the pattern for calculating the amino acid profile of other similar foods. The amino acid values for the pattern are expressed on a per-gram-of-nitrogen basis. Amino acids are extracted in three groups—tryptophan, sulfur-containing amino acids (methionine and cystine), and all others. Tryptophan is determined by alkaline hydrolysis/HPLC (AOAC 988.15), methionine and cystine by performic oxidation/HPLC (AOAC 994.12) and all others by acid hydrolysis/HPLC (AOAC 982.30). Hydroxyproline in meats has been determined using a colorimetric method (AOAC 990.26). The amino acid patterns and the total nitrogen content are used to calculate the levels of individual amino acids per 100 g of food, using the following formula:

$$AA_f = (AA_n * V_p)/N_f$$

where:
AA_f = amino acid content per 100 g of food,
AA_n = amino acid content per g of nitrogen,
V_p = protein content of food, and
N_f = nitrogen factor.

For foods processed in the NDBS since SR14 (2001), the number of observations used in developing an amino acid pattern is released only with the pattern. The amino acid profiles calculated from these patterns will show the number of data points to be zero. In the past, the number of data points appeared only for the food item for which the amino

acid pattern was developed, not on other foods that used the same pattern. It referred to the number of observations used in developing the amino acid pattern for that food.

If amino acid values are presented for an item with more than one protein-containing ingredient, the values may be calculated on a per-gram-of-nitrogen basis from the amino acid patterns of the various protein-containing ingredients. Then the amino acid contents for an item on the 100-g basis are calculated as the sum of the amino acids in each protein-containing ingredient multiplied by total nitrogen in the item. The number of data points for these values is given as zero.

Weights and Measures

Information is provided on household measures for food items (for example, 1 cup, 1 tablespoon, 1 fruit, 1 leg). Weights are given for edible material without refuse, that is, the weight of an apple without the core or stem, or a chicken leg without the bone, and so forth. The Weight file contains the gram weights and measure descriptions for each food item. This file can be used to calculate nutrient values for food portions from the values provided per 100 g of food. The following formula is used to calculate the nutrient content per household measure:

$$N = (V*W)/100$$

where:

N = nutrient value per household measure,
V = nutrient value per 100 g (Nutr_Val in the Nutrient Data file), and
W = g weight of portion (Gm_Wgt in the Weight file).

The Weight file can be used to produce reports showing the household measure and nutrient values calculated for that portion. The weights are derived from published sources, industry files, studies conducted by USDA (Adams, 1975; Fulton *et al*., 1977), and the weights and measures used in the FNDDS (2012). However, weight information is not available for all food items in the database. Though special efforts have been made to provide representative values, weights and measures obtained from different sources vary considerably for some foods. The format of this file is described on p. 36.

Footnotes

Footnotes are provided for a few items where information about food description, weights and measures, or nutrient values could not be accommodated in existing fields. For example, if citric acid is added to a juice drink, this is indicated in the footnote. The format of this file is described on p. 36.

Sources of Data

The Sources of Data file (previously called References) was first added with SR14 (2001). The name of the file and fields reflect the fact that not all sources are journals or published literature, but also include the results of unpublished data from USDA-sponsored research and from research sponsored by others either separately or in collaboration with USDA. It contains data sources for the nutrient values and links to an identification number on each nutrient record. Nutrient-specific source documentation is not electronically available for data added prior to SR14 (2001). Data source information for these food items will be added when new data are generated and published in future releases. The format of this file is described on p. 37.

The Sources of Data Link file is provided to allow users to establish a relationship between the Sources of Data file and the Nutrient Data file. This lets the user identify specific sources of data for each nutrient value. For example, the user can use these files to determine the dates associated with source documents for a particular data value. These files can also be used to determine values obtained from a particular data source, for example where NFNAP data is used in the database. The format of this file is described on p.37.

Explanation of File Formats

The data appear in two different organizational formats. One is a relational format of four principal and six support files making up the database. The relational format is complete and contains all food, nutrient, and related data. The other is a flat abbreviated file with all the food items, but fewer nutrients, and not all of the other related information. The abbreviated file does not include values for starch, individual sugars, fluoride, betaine, vitamin D_2 or D_3, added vitamin E, added vitamin B_{12}, alcohol, caffeine, theobromine, phytosterols, individual amino acids, or individual fatty acids. See p. 38 for more information on this file.

Relational Files

The four principal database files are the Food Description file, Nutrient Data file, Gram Weight file, and Footnote file. The eight support files are the Food Group Description file, LanguaL Factor file, LanguaL Factor Description file, Nutrient Definition file, Source Code file, Data Derivation Code Description file, Sources of Data file, and Sources of Data Link file. Table 3 shows the number of records in each file. In a relational database, these files can be linked together in a variety of combinations to produce queries and generate reports. Figure 1 provides a diagram of the relationships between files and their key fields.

The relational files are provided in both ASCII (ISO/IEC 8859-1) format and a Microsoft Access 2007 database. Tables 4 through 13 describe the formats of these files. Information on the relationships that can be made among these files is also given.

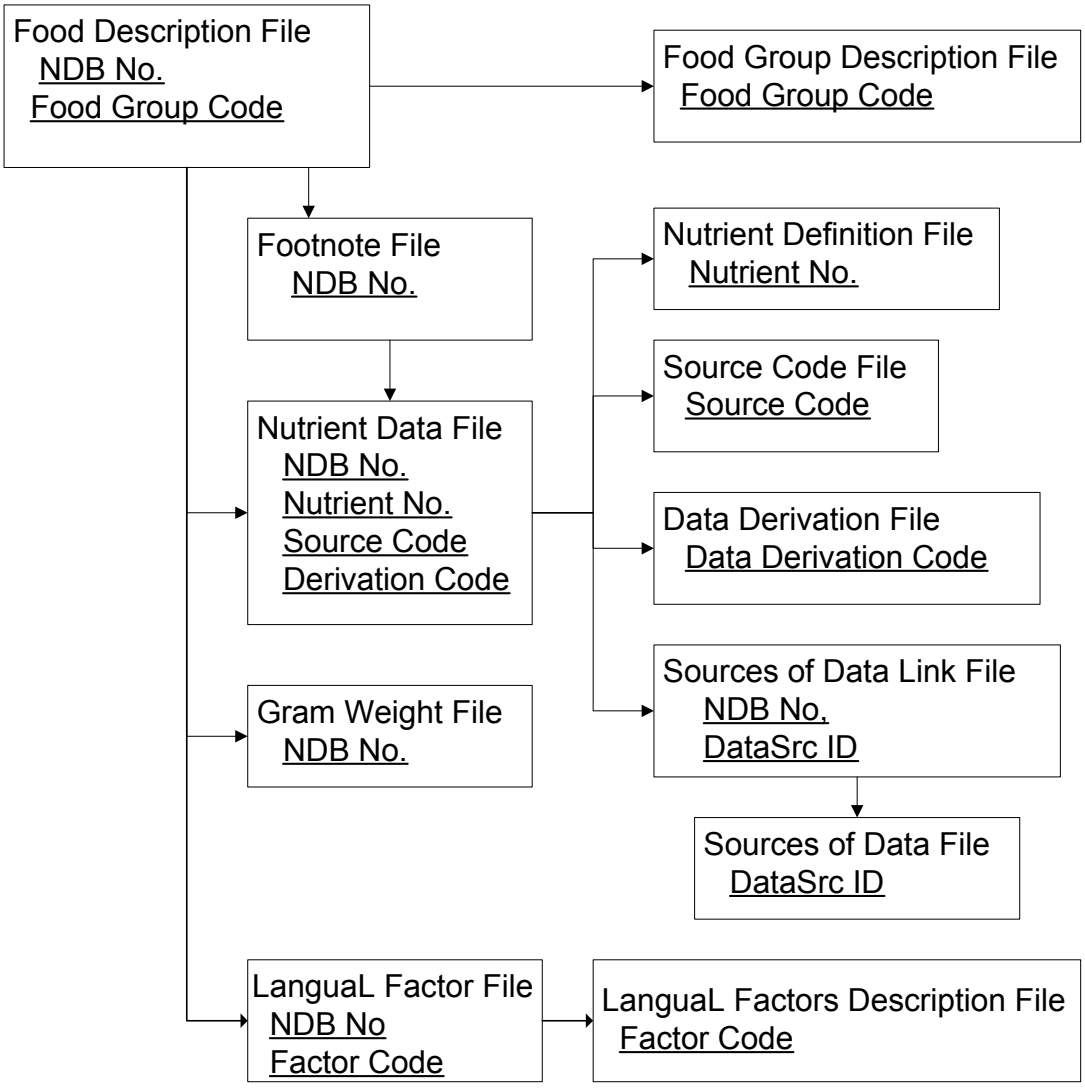
Fields that always contain data and fields that can be left blank or null are identified in the "blank" column; N indicates a field that is always filled; and Y indicates a field that may be left blank (null) (Tables 4-13). An asterisk (*) indicates primary key(s) for the file. Though keys are not identified for the ASCII files, the file descriptions show where keys are used to sort and manage records within the NDBS. When importing these files into a database management system, if keys are to be identified for the files, it is important to use the keys listed here, particularly with the Nutrient Data file, which uses two.

Table 3. – Number of Records in Principal and Support Files

File name	Table name	Number of records
Principal files		
Food Description (p. 29)	FOOD_DES	8,618
Nutrient Data (p. 32)	NUT_DATA	654,572
Weight (p. 36)	WEIGHT	15,228
Footnote (p. 36)	FOOTNOTE	531
Support files		
Food Group Description (p. 31)	FD_GROUP	25
LanguaL Factor (p. 31)	LANGUAL	38,725
LanguaL Factors Description (p. 31)	LANGDESC	774
Nutrient Definition (p. 34)	NUTR_DEF	150
Source Code (p. 34)	SRC_CD	10
Data Derivation Code Description (p. 35)	DERIV_CD	55
Sources of Data (p. 37)	DATA_SRC	655
Sources of Data Link (p. 37)	DATSRCLN	230,663

ASCII files are delimited. All fields are separated by carets (^) and text fields are surrounded by tildes (~). A double caret (^^) or two carets and two tildes (~~) appear when a field is null or blank. Format descriptions include the name of each field, its type [N = numeric with width and number of decimals (w.d) or A = alphanumeric], and maximum record length. The actual length in the data files may be less and most likely will change in later releases. Values will be padded with trailing zeroes when imported into various software packages, depending on the formats used.

Figure 1. Relationships among files in the USDA National Nutrient Database for Standard Reference *



* Underlined items denote key fields.

Food Description File (file name = FOOD_DES). This file (Table 4) contains long and short descriptions and food group designators for all food items, along with common names, manufacturer name, scientific name, percentage and description of refuse, and factors used for calculating protein and kilocalories, if applicable. Items used in the FNDDS are also identified by value of "Y" in the Survey field.

- Links to the Food Group Description file by the FdGrp_Cd field
- Links to the Nutrient Data file by the NDB_No field
- Links to the Weight file by the NDB_No field
- Links to the Footnote file by the NDB_No field
- Links to the LanguaL Factor file by the NDB_No field

Table 4.—Food Description File Format

Field name	Type	Blank	Description
NDB_No	A 5*	N	5-digit Nutrient Databank number that uniquely identifies a food item. If this field is defined as numeric, the leading zero will be lost.
FdGrp_Cd	A 4	N	4-digit code indicating food group to which a food item belongs.
Long_Desc	A 200	N	200-character description of food item.
Shrt_Desc	A 60	N	60-character abbreviated description of food item. Generated from the 200-character description using abbreviations in Appendix A. If short description is longer than 60 characters, additional abbreviations are made.
ComName	A 100	Y	Other names commonly used to describe a food, including local or regional names for various foods, for example, “soda” or “pop” for “carbonated beverages.”
ManufacName	A 65	Y	Indicates the company that manufactured the product, when appropriate.
Survey	A 1	Y	Indicates if the food item is used in the USDA Food and Nutrient Database for Dietary Studies (FNDDS) and thus has a complete nutrient profile for the 65 FNDDS nutrients.
Ref_desc	A 135	Y	Description of inedible parts of a food item (refuse), such as seeds or bone.
Refuse	N 2	Y	Percentage of refuse.
SciName	A 65	Y	Scientific name of the food item. Given for the least processed form of the food (usually raw), if applicable.
N_Factor	N 4.2	Y	Factor for converting nitrogen to protein (see p. 12).
Pro_Factor	N 4.2	Y	Factor for calculating calories from protein (see p. 13).
Fat_Factor	N 4.2	Y	Factor for calculating calories from fat (see p. 13).
CHO_Factor	N 4.2	Y	Factor for calculating calories from carbohydrate (see p. 13).

* Primary key for the Food Description file.

Food Group Description File (file name = FD_GROUP). This file (Table 5) is a support file to the Food Description file and contains a list of food groups used in SR27 and their descriptions.

- Links to the Food Description file by FdGrp_Cd

Table 5.—Food Group Description File Format

Field name	Type	Blank	Description
FdGrp_Cd	A 4*	N	4-digit code identifying a food group. Only the first 2 digits are currently assigned. In the future, the last 2 digits may be used. Codes may not be consecutive.
FdGrp_Desc	A 60	N	Name of food group.

* Primary key for the Food Group Description file.

***LanguaL Factor* File** (File name = LANGUAL). This file (Table 6) is a support file to the Food Description file and contains the factors from the LanguaL Thesaurus used to code a particular food.

- Links to the Food Description file by the NDB_No field
- Links to LanguaL Factors Description file by the Factor_Code field

Table 6.—LanguaL Factor File Format

Field name	Type	Blank	Description
NDB_No	A 5*	N	5-digit Nutrient Databank number that uniquely identifies a food item. If this field is defined as numeric, the leading zero will be lost.
Factor_Code	A 5*	N	The LanguaL factor from the Thesaurus

* Primary key for the LanguaL Factor file.

LanguaL Factors Description File (File name = LANGDESC). This file (Table 7) is a support file to the LanguaL Factor file and contains the descriptions for only those factors used in coding the selected food items codes in this release of SR.

- Links to the LanguaL Factor File by the Factor_Code field

Table 7.—LanguaL Factors Description File Format

Field name	Type	Blank	Description
Factor_Code	A 5*	N	The LanguaL factor from the Thesaurus. Only those codes used to factor the foods contained in the LanguaL Factor file are included in this file
Description	A 140	N	The description of the LanguaL Factor Code from the thesaurus

* Primary key for the LanguaL Factor Description file.

Nutrient Data File (file name = NUT_DATA)**.** This file (Table 8) contains the nutrient values and information about the values, including expanded statistical information.

- Links to the Food Description file by NDB_No
- Links to the Food Description file by Ref_NDB_No
- Links to the Weight file by NDB_No
- Links to the Footnote file by NDB_No and when applicable, Nutr_No
- Links to the Nutrient Definition file by Nutr_No
- Links to the Source Code file by Src_Cd
- Links to the Derivation Code file by Deriv_Cd

Table 8.—Nutrient Data File Format

Field name	Type	Blank	Description
NDB_No	A 5*	N	5-digit Nutrient Databank number.
Nutr_No	A 3*	N	Unique 3-digit identifier code for a nutrient.
Nutr_Val	N 10.3	N	Amount in 100 grams, edible portion †.
Num_Data_Pts	N 5.0	N	Number of data points (previously called Sample_Ct) is the number of analyses used to calculate the nutrient value. If the number of data points is 0, the value was calculated or imputed.
Std_Error	N 8.3	Y	Standard error of the mean. Null if cannot be calculated. The standard error is also not given if the number of data points is less than three.
Src_Cd	A 2	N	Code indicating type of data.
Deriv_Cd	A 4	Y	Data Derivation Code giving specific information on how the value is determined. This field is populated only for items added or updated starting with SR14.

Field name	Type	Blank	Description
Ref_NDB_No	A 5	Y	NDB number of the item used to calculate a missing value. Populated only for items added or updated starting with SR14.
Add_Nutr_Mark	A 1	Y	Indicates a vitamin or mineral added for fortification or enrichment. This field is populated for ready-to-eat breakfast cereals and many brand-name hot cereals in food group 8.
Num_Studies	N 2	Y	Number of studies.
Min	N 10.3	Y	Minimum value.
Max	N 10.3	Y	Maximum value.
DF	N 4	Y	Degrees of freedom.
Low_EB	N 10.3	Y	Lower 95% error bound.
Up_EB	N 10.3	Y	Upper 95% error bound.
Stat_cmt	A 10	Y	Statistical comments. See definitions below.
AddMod_Date	A10	Y	Indicates when a value was either added to the database or last modified.
CC	A 1	Y	Confidence Code indicating data quality, based on evaluation of sample plan, sample handling, analytical method, analytical quality control, and number of samples analyzed. Not included in this release, but is planned for future releases.

* Primary keys for the Nutrient Data file.
† Nutrient values have been rounded to a specified number of decimal places for each nutrient. Number of decimal places is listed in the Nutrient Definition file.

Definitions of each statistical comment included in the Nutrient Data table follow:

1. The displayed summary statistics were computed from data containing a verbally described value, e.g., less than, trace or not-detected. This must be done in order to carry out any mathematical computations. In every case, the definition of the verbal value is used to derive a numeric value. By definition the actual verbal definition must be a value that falls between two or more numeric values, using a simple linear interpolation between these values. This will derive the most likely numeric location for the verbally described value by following simplest and least computationally intense imputation procedures. If there is a need to account for the added variance due to the imputation of the summary value, see Little and Rubin (2002). Further information is provided in Appendices D and E.

2. The displayed degrees of freedom were computed using Satterthwaite's approximation (Kotz and Johnson, 1988).
3. The procedure used to estimate the reliability of the generic mean requires that the data associated with each study be a simple random sample from all the products associated with the given data source (for example, manufacturer, variety, cultivar, and species). For this nutrient, one or more data sources had only one observation. Therefore, the standard errors, degrees of freedom, and error bounds were computed from the between-group standard deviation of the weighted groups having only one observation.

Nutrient Definition File (file name = NUTR_DEF). This file (Table 9) is a support file to the Nutrient Data file. It provides the 3-digit nutrient code, unit of measure, INFOODS tagname, and description.

- Links to the Nutrient Data file by Nutr_No

Table 9.—Nutrient Definition File Format

Field name	Type	Blank	Description
Nutr_No	A 3*	N	Unique 3-digit identifier code for a nutrient.
Units	A 7	N	Units of measure (mg, g, µg, and so on).
Tagname	A 20	Y	International Network of Food Data Systems (INFOODS) Tagnames.† A unique abbreviation for a nutrient/food component developed by INFOODS to aid in the interchange of data.
NutrDesc	A 60	N	Name of nutrient/food component.
Num_Dec	A 1	N	Number of decimal places to which a nutrient value is rounded.
SR_Order	N 6	N	Used to sort nutrient records in the same order as various reports produced from SR.

* Primary key for the Nutrient Definition file.
† INFOODS, 2014.

Source Code File (file name = SRC_CD). This file (Table 10) contains codes indicating the type of data (analytical, calculated, assumed zero, and so on) in the Nutrient Data file. To improve the usability of the database and to provide values for the FNDDS, NDL staff imputed nutrient values for a number of proximate components, total dietary fiber, total sugar, and vitamin and mineral values.

- Links to the Nutrient Data file by Src_Cd